

BAO–HHT Integration Update

Generated: 2025-10-09 19:27

Combined Fit (BAO + Pantheon+SH0ES)

BAO Helper Check @ Combined Best-fit

chi2_diag: 243.34156390594492

chi2_full: 236.4174852773252

perz_sum_full: 236.417485278288

N: 14

rho: None

Analytic ρ^* at Combined Best-fit

rho_star: 0.17808558892454637

sigma_rho: 0.06389570370941773

chi2_at_rho_star: 237.16770419356172

Per-z 2x2 DM–DH χ^2 (full covariance)

z	pDM	pDH	rho	chi2_pair_di ag	chi2_pair_fu ll
0.38	7.77504908 2519165	5.38091698 7970684	0.17399812 65976199	89.4056558 670136	77.1833256 6635239
0.51	8.32804969 43531	1.25463453 71043806	0.17399812 66006928	70.9305195 3330988	69.3953951 7186432
0.7	7.50996127 4674717	-1.99072788 86261963	0.17399812 66018801	60.3625158 73668016	67.6121416 7482791
0.845	-3.23340158 61341063	-1.43666669 6244528	0.17399812 65988715	12.5188970 1331272	11.2427201 95173394
0.85	-0.49722630 07160289	1.24433939 74000296	0.17399812 65989746	1.79561453 0045616	2.07370841 57326127
1.48	2.16109554 09859596	-0.52804574 93259949	0.17399812 65987391	4.94916625 0650649	5.51319873 0837014
2.33	-1.89684795 45686967	-1.06252184 0686581	0.17399812 66032101	4.72698482 4687449	4.15130049 522819

Per-point BAO residuals (obs, pred, σ , pull, χ^2_i)

i	z	kind	obs	pred	sigma	pull	chi2_i
0	0.38	DH_over _rd	24.8997	23.6792	0.22682	5.38091	28.9542
			9828875	6772592	5755824	6987970	6763143
			624	9055	8426	684	15
1	0.38	DM_over _rd	10.2653	9.81388	0.05806	7.77504	60.4513
			3360631	0388245	4356028	9082519	8823558
			0069	538	2525	165	211
2	0.51	DH_over _rd	22.4196	22.1858	0.18637	1.25463	1.57410
			7458084	4520127	2503433	4537104	7821695
			5346	046	9874	3806	1232
3	0.51	DM_over _rd	13.3777	12.7948	0.06999	8.32804	69.3564
			7814018	6219888	4291903	9694353	1171161
			1809	576	8175	1	475
4	0.7	DH_over _rd	19.7294	20.0830	0.17759	-1.99072	3.96299
			7439714	1334523	2804176	7888626	7526554
			1736	54	5879	1963	1137
5	0.7	DM_over _rd	17.6757	16.8083	0.11549	7.50996	56.3995
			2384176	4284154	7399853	1274674	1834711
			586	2125	2773	717	3904
6	0.845	DH_over _rd	16.5427	18.5828	1.41999	-1.43666	2.06401
			7616398	3906536	7349919	6696244	1196098
			0086	4244	0297	528	167
7	0.845	DM_over _rd	18.33	19.6103	0.39597	-3.23340	10.4548
				6170519	9797464	1586134	8581721
				8667	4666	1063	4556
8	0.85	DH_over _rd	19.6	18.5330	0.85747	1.24433	1.54838
				0795732	6702017	9397400	0535921
				715	3607	0296	869
9	0.85	DM_over _rd	19.5	19.7031	0.40856	-0.49722	0.24723
				5126826	9031790	6300716	3994123
				4083	0235	0289	7468
10	1.48	DH_over _rd	13.2332	13.3290	0.18143	-0.52804	0.27883
			6449317	6801973	0352719	5749325	2313381
			8956	1046	2026	9949	2515
11	1.48	DM_over _rd	30.2774	29.6293	0.29988	2.16109	4.67033
			4706146	7268287	2335741	5540985	3937269
			7285	6085	403	9596	3975
12	2.33	DH_over _rd	8.93	9.04264	0.10601	-1.06252	1.12895
				6450345	8009260	1840686	2661936
				243	3606	581	0004
13	2.33	DM_over _rd	37.6	38.9631	0.71863	-1.89684	3.59803
				3252152	0356346	7954568	2162751
				7385	7905	6967	4487

Leave-one-z-pair-out $\Delta\chi^2$

z_omitted	chi2_full	chi2_leave1out	delta
0.38	237.171790314201	159.988464663544	-77.183325650657

	3	14	17
0.51	237.171790314201 3	167.776395156376 48	-69.395395157824 82
0.7	237.171790314201 3	169.559648645198 08	-67.612141669003 22
0.845	237.171790314201 3	225.929070119103 4	-11.242720195097 888
0.85	237.171790314201 3	235.098081898473 25	-2.0737084157280 49
1.48	237.171790314201 3	231.658591583469 16	-5.5131987307321 42
2.33	237.171790314201 3	233.020489819043 3	-4.1513004951580 115

Notes & Implications

- BAO-only prefers $\rho \approx -0.118 \pm 0.129$ with $\chi^2 \approx 59.60$ (at $H_0 \approx 68.73$, $\Omega_m \approx 0.293$).
- SN-anchored ($H_0 \approx 73.12$, $\Omega_m \approx 0.232$) yields $\rho^* \approx +0.143 \pm 0.066$ and BAO $\chi^2 \approx 226.53$.
- Combined best-fit prefers $\rho^* \approx +0.174 \pm 0.064$; BAO $\chi^2 \approx 236-237$ consistent with Λ .
- LOOCV shows low- z pairs ($z \approx 0.38, 0.51, 0.70$) dominate χ^2 ; extended- z testing should check stability of ρ across z and the influence of those pairs.

Recommended Next Steps

- 1) Test a 2-bin $\rho(z)$: $z < 0.9$ vs $z \geq 0.9$. Compare AIC/BIC to constant- ρ .
- 2) Do targeted LOOCV/HHT slices around $z \approx 0.38-0.70$ to probe systematics.
- 3) If adding higher- z BAO, rerun fit- ρ and export-residuals for the augmented set; check whether ρ^* and per- z χ^2 stabilize.
- 4) If needed, generalize to $\rho(z)$ with one extra d.o.f. (e.g., 2-bin or weak linear), but guard against overfitting ($\Delta\text{AIC}/\Delta\text{BIC}$).